Kingdom of Saudi Arabia
Ministry of Education
KSA

GRADUATION PROJECT · IT DEPARTMENT

Prediction of Emergency Cases Application Usage

A Security-Trust Perspective

Asma Almurjan | Baida Alfawzan | Jori Alkhuwailed | Reema Aljodaye

Supervisor: **Prof. Ali Alkhalifah**

B.Sc. in Information Technology · Qassim University, Saudi Arabia

ABSTRACT

Emergency mobile applications are critical tools for saving lives during crises. Despite their potential, user adoption remains low due to concerns about security, privacy, and trust. This project develops a predictive model that evaluates user trust and security perceptions toward emergency applications, identifying key factors influencing usage behavior. The study employs machine learning techniques (Scikit-learn, TensorFlow) with survey data to analyze perceived security, privacy, usability, and user experience. Results indicate that transparency in data handling, encryption features, and user-friendly design significantly enhance trust and adoption rates.

Emergency Apps · User Trust · Security Perception · Machine Learning · Predictive Modeling · Privacy

PROBLEM & MOTIVATION

Despite Saudi Arabia's investments in digital emergency infrastructure (e.g., SAFEER app), user engagement remains below expected levels. Key barriers include:

- Security concerns: data breaches & unauthorized tracking
- Privacy dilemma: sensitive data access deters users
- Trust deficit: lack of data usage transparency
- Cultural factors: local data hosting & Arabic UI needs

GOALS & OBJECTIVES

1. Design multi-scenario disaster prediction platform
2. Integrate ML algorithms (LSTM, Random Forest, Isolation Forest)
3. Implement trust evaluation framework
4. Validate through structured simulation drills
5. Provide actionable disaster recovery plans (DRP)

METHODOLOGY

Waterfall model with mixed-methods approach:

Requirements & Risk › Strategy Design › Implementation › Testing & Validation

Data collection: Mixed-methods · Survey (300-400 participants) · ML prediction

Python · Scikit-learn · TensorFlow · HTML/CSS/JS · IndexedDB

EMERGENCY SCENARIOS

Platform covers four IT disaster scenarios:

- Network Outage Prediction
- Database Failure Forecasting
- Cyber Attack Anticipation
- Insider Threat Detection

Risk scoring: Low (<45) · Medium (45-69) · High (≥70) out of 100

RESULTS & FINDINGS

Key trust enhancement factors:

Factor	Level
Transparency	High
Encryption	High
UI Design	High
Local Data	High

95% Objectives achieved · **27** Tests passed · **78%** Avg quiz score

CONCLUSION & FUTURE WORK

The Emergency AI System was successfully deployed as a web-based platform for risk prediction, simulation, and security awareness — with client-side IndexedDB privacy-focused storage.

System contributions:

- ✓ Six-step prediction wizard with AI recommendations
- ✓ Admin & user dashboards with incident reporting
- ✓ Interactive security awareness quiz

Future work:

Integrating real ML models (LSTM, Random Forest), cloud-based database migration, real-time push notifications, and Arabic language interface.